

Final Evaluation: Project 1

GANFORGE - Image Style Transfer

Introduction to the Project

What is Style Transfer?

Applying the artistic style (like brushstrokes and textures) of one image to the content structure of another.

Why is it exciting?

It beautifully merges artificial intelligence with art. It has practical applications in gaming, filters, and film post-production.

Cycle-GAN Architecture

Cycle-Consistent Generative Adversarial Networks (Cycle-GANs) allow for image-to-image translation without paired data.

It works by utilizing two generators and two discriminators, passing the image back and forth to ensure consistency (Cycle Consistency Loss).

Example: Turning a picture of a horse into a zebra, and then converting that zebra back into the exact same horse.

Loss Functions Explained

The Loss Function combines multiple parts:

- 1. Adversarial Loss:** Ensures the generated images look incredibly real, trying to fool the discriminator.
- 2. Cycle Consistency Loss:** Ensures that structure and content are preserved during the translation process.

Conclusion & Results

We successfully transformed ordinary, everyday photographs into beautiful artwork mimicking the style of Claude Monet.

This project clearly demonstrates the capability of Cycle-GANs to perform high-quality artistic style transfer without paired datasets.

Thank you!